H 3 - Momentum - role owner

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a) Checking the conditions (α) - (e) at $x = (0,5; 0,5; 0,5)$

$\pi = (2, 3, 3)$

(α) Primal feasibility. All three Rows: $0,5 + 0,5 = 1 = b_i$ exactly.

Bounds $0 \leq 0,5 \leq 1$. Max violation $= 0$

(β) Dual feasibility (signs). All three rows are "$\geq$",

need $\pi_i \geq 0$: $2, 3, 3 \geq 0$, Max violation $= 0$

(c) Complementary slackness. Every row is tight

(slack $= 1 - 1 = 0$ for all three), so $\pi_i \cdot \text{slack}_i = \pi_i \cdot 0 = 0$

for every i. Max $= 0$

(d) Reduced costs. $r_j = c_j - \sum_i \pi_i A_{ij}$:

$r_1 = 5 - (2 \cdot 1 + 3 \cdot 0 + 3 \cdot 1) = 5 - 5 = 0$

$r_2 = 5 - (2 \cdot 1 + 3 \cdot 1 + 3 \cdot 0) = 5 - 5 = 0$

$r_3 = 6 - (2 \cdot 0 + 3 \cdot 1 + 3 \cdot 1) = 6 - 6 = 0$

All three $y_j = 0,5$ are strictly interior ($0 < y_j < 1$), so

each needs $r_j = 0$ - satisfied exactly. Max violation $= 0$.

(e) Strong duality. $c^T x = 8$ $b^T \pi = 1 \cdot 2 + 1 \cdot 3 + 1 \cdot 3 = 8$

Sum of $\min(0, r_j)$ over the three (upper-Bounded by - 1) var:

$\min(0, 0) + \min(0, 0) + \min(0, 0) = 0$. So $b^T \pi + (\text{Sum}) =$

$8 + 0 = 8 = c^T x$ difference $= 0$.

All five numbers are exactly $0 - x = (0,5; 0,5; 0,5)$

together with $\pi = (2, 3, 3)$ is genuine, fully

certified optimal primal - dual pair.

B) With $\pi' = (3, 2, 2)$: which conditions fail, and by how much? Which π we pick

Conditions (a), (B), (c) are unchanged by which π we pick

Recomputing (d):

$r_1' = 5 - (3 \cdot 1 + 2 \cdot 0 + 2 \cdot 1) = 5 - 5 = 0$

$r_2' = 5 - (3 \cdot 1 + 2 \cdot 1 + 2 \cdot 0) = 5 - 5 = 0$

$r_3' = 6 - (3 \cdot 0 + 2 \cdot 1 + 2 \cdot 1) = 6 - 4 = 2$

Since $0 < y_3 < 1$ needs $r_3' = 0$, condition (d) fails by 2

Then (e): $b^T \pi' = 1 \cdot 3 + 1 \cdot 2 + 1 \cdot 2 = 7$

$\sum \min(0, r_j') = 0 + 0 + \min(0, 2) = 0$; so the dual value is $7 + 0 = 7 \neq 8 = c^T x$. Condition (e) fails by a gap of 1

(f) π' still bounds z_{LP} from below. Weak duality for a box-constrained LP says that for any sign-feasible π (condition (B) only, not needing (d) or (e)), the value $b^T \pi + \sum_j \min(0, r_j)$ is a valid lower bound on z_{LP} - it doesn't require π to be optimal dual vector. Here that value is $7 \leq z_{LP} = 8$: still a correct, if weaker, bound. It is 1 below the best (tight) bound of 8 achieved by the true dual $\pi = (2, 3, 3)$ - exactly the duality-gap number found in (B)